HDOC DAY – 73

Algorithm

Main Algorithm

1. Start.
2. Define an enumeration Days containing SUNDAY to SATURDAY.
3. Define another enumeration TrafficLight containing RED, YELLOW, and GREEN.
4. Display the menu:
    - Display days and their integer values.
    - Display traffic-light action.
5. Read the user's choice.
6. If choice is 1, call displayDays().
7. If choice is 2, call trafficSignal().
8. Otherwise, display "Invalid choice".
9. Stop.

Algorithm for displayDays()

1. Start the function.
2. Print each day along with its enumeration integer value.
3. Since no values are assigned manually, enumeration starts from 0.
4. Therefore, SUNDAY = 0 through SATURDAY = 6.
5. Return to main().

Algorithm for trafficSignal()

1. Start the function.
2. Display RED, YELLOW, and GREEN.
3. Read the traffic-light choice.
4. Assign the choice to the enumeration variable.
5. Using switch:
    - RED → Display "Stop".
    - YELLOW → Display "Wait".

- GREEN → Display "Go".

6. For any other value, display "Invalid traffic light".
7. Return to main().

C PROGRAM:

```
#include <stdio.h>

/* Enumeration for days */
enum Days
{
   SUNDAY,
   MONDAY,
   TUESDAY,
   WEDNESDAY,
   THURSDAY,
   FRIDAY,
   SATURDAY
};

/* Enumeration for traffic lights */
enum TrafficLight
{
   RED,
   YELLOW,
   GREEN
};

void displayDays()
{
```

```
    printf("SUNDAY = %d\n", SUNDAY);
    printf("MONDAY = %d\n", MONDAY);
    printf("TUESDAY = %d\n", TUESDAY);
    printf("WEDNESDAY = %d\n", WEDNESDAY);
    printf("THURSDAY = %d\n", THURSDAY);
    printf("FRIDAY = %d\n", FRIDAY);
    printf("SATURDAY = %d\n", SATURDAY);
}

void trafficSignal()
{
    enum TrafficLight light;
    int choice;

    printf("\nEnter traffic light:\n");
    printf("0. RED\n");
    printf("1. YELLOW\n");
    printf("2. GREEN\n");
    printf("Enter choice: ");
    scanf("%d", &choice);

    light = choice;

    switch (light)
    {
        case RED:
            printf("Stop\n");
            break;
```

```
        case YELLOW:
            printf("Wait\n");
            break;

        case GREEN:
            printf("Go\n");
            break;

        default:
            printf("Invalid traffic light\n");
    }
}

int main()
{
    int choice;

    printf("1. Display days with integer values\n");
    printf("2. Display traffic light action\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            displayDays();
            break;
```

```
        case 2:
            trafficSignal();
            break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}
```

TEST CASE TABLE:

Test Case	Menu Choice	Traffic Light	Expected Output
1	1	—	SUNDAY = 0 through SATURDAY = 6
2	2	RED (0)	Stop
3	2	YELLOW (1)	Wait
4	2	GREEN (2)	Go
5	2	Invalid value	Invalid traffic light